Chapter 10

Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\boldsymbol{A}$

$$\boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

$$\left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

can be written in the more symmetrical form

$$\Box^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0}\rho$$

$$\Box^2\boldsymbol{A} - \boldsymbol{\nabla}L = -\mu_0\boldsymbol{J}$$

where

$$\Box^2 \equiv \boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\Box^2$ and L into the equations for V:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)V + \frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \boldsymbol{\nabla}^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) + \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} =$$

$$= \boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

and for $\boldsymbol{A}$:

$$\left(\nabla^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} - \nabla\left(\nabla\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \left(\nabla^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \nabla\left(\nabla\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

□

Problem 10.2

Solution:

a) The potentials are given as

$$V = 0, \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c}(ct - |x|)^2\hat{\mathbf{z}} \text{ for } |x| < ct \\ \mathbf{0} \qquad\qquad\qquad\quad \text{for } |x| > ct \end{cases}$$

with the fields

$$\mathbf{E} = -\frac{\mu_0 k}{2}(ct - |x|)\hat{\mathbf{z}}$$

$$\mathbf{B} = \frac{\mu_0 k}{2c}(ct - |x|)\hat{\mathbf{y}}$$

At time t_1, the vector magnetic potential is $\mathbf{0}$ because

$$|x| \geq ct_1 \text{ gives } |d| \geq c\frac{d}{c} \rightarrow d \geq d$$

and thus no energy in the box at this time.

At time t_2 however, the potential has reached the entire box

The energy is given by

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right)$$

and since

$$B = \frac{E}{c}, c^2 = \frac{1}{\mu_0\epsilon_0}$$

we get

$$u = \frac{1}{2}\left(\epsilon_0 + \frac{1}{\mu_0 c^2}\right)E^2 = \epsilon_0 E^2$$

The electric field squared at time t_2 is

$$E^2 = \frac{\mu_0^2 k^2}{4}(ct_2 - |x|)^2 = \frac{\mu_0^2 k^2}{4}(d + h - x)^2 \quad (x \text{ always positive})$$

Integrating over the rectangular box in the $\hat{\mathbf{x}}$-direction (since the electric field is constant in $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$, so we multiply the result by wl), we get

$$lw\int_d^{d+h} \epsilon_0 E^2 \,\mathrm{d}x = \frac{\epsilon_0 \mu_0^2 k^2 lw}{4}\int_d^{d+h}(d+h-x)^2\,\mathrm{d}x = \frac{\epsilon_0\mu_0^2k^2lw}{4}\left[-\frac{(d+h-x)^3}{3}\right]_d^{d+h} =$$

$$= \frac{\epsilon_0\mu_0^2k^2lwh^3}{12}$$

b) The Poynting vector is given by

$$\mathbf{S} = \frac{1}{\mu_0}\mathbf{E}\times\mathbf{B}$$

With $\mathbf{E} = E(-\hat{\mathbf{z}})$ and $\mathbf{B} = B\hat{\mathbf{y}}$, the cross product becomes

$$EB\hat{\mathbf{x}}$$

which gives

$$\mathbf{S} = \frac{1}{\mu_0}\frac{\mu_0^2k^2}{4c}\hat{\mathbf{x}} = \frac{\mu_0 k^2}{4c}(ct - |x|)^2\hat{\mathbf{x}}$$

The energy flowing in is the integral of the Poynting vector at $x = d$

$$P = \int_S \mathbf{S}\cdot\mathrm{d}\boldsymbol{a} = \frac{\mu_0 k^2 lw}{4c}(ct-d)^2$$

c)

$$W = \int_{d/c}^{(d+h)/c} P\,\mathrm{d}t = \frac{\mu_0 k^2 lw}{4c}\int_{d/c}^{(d+h)/c}(ct-d)^2\,\mathrm{d}t =$$

$$= \frac{\mu_0 k^2 lw}{4c}\left[\frac{(ct-d)^3}{3c}\right]_{d/c}^{(d+h)/c} = \frac{\mu_0 k^2 lw}{4c}\left(\frac{(d+h-d)^3}{3c} - \frac{(d-d)^3}{3c}\right) =$$

$$= \frac{\mu_0 k^2 lwh^3}{12c^2} = \frac{\epsilon_0\mu_0^2k^2lwh^3}{12}$$

□

Problem 10.3

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\boldsymbol{r},t)=0,\quad \boldsymbol{A}(\boldsymbol{r},t)=-\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

(b) Use the gauge function $\lambda=-(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Solution:

The **E** and **B** fields are

$$\mathbf{E}=-\boldsymbol{\nabla}V-\frac{\partial\mathbf{A}}{\partial t}$$
$$\mathbf{B}=\boldsymbol{\nabla}\times\mathbf{A}$$

The derivative of **A** with respect to time is

$$\frac{\partial\mathbf{A}}{\partial t}=-\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}},$$

while the curl of **A** in spherical coordinates is

$$\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta A_\phi)-\frac{\partial A_\theta}{\partial\phi}\right]\hat{\mathbf{r}}+\frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi}-\frac{\partial}{\partial r}(rv_\phi)\right]\hat{\boldsymbol{\theta}}+\frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta)-\frac{\partial A_r}{\partial\theta}\right]\hat{\boldsymbol{\phi}}.$$

Since $A_\phi=A_\theta=0$, and $\frac{\partial A_r}{\partial\phi}=\frac{\partial A_r}{\partial\theta}=0$, the curl of **A** is zero.

(a)

$$\mathbf{E}=-\frac{\partial\mathbf{A}}{\partial t}=\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$
$$\mathbf{B}=\boldsymbol{\nabla}\times\mathbf{A}=0$$

(b) Using the gauge function, V and **A** are transformed into

$$V'(\mathbf{r},t)=V(\mathbf{r},t)-\frac{\partial\lambda}{\partial t}$$
$$\mathbf{A}'(\mathbf{r},t)=\mathbf{A}(\mathbf{r},t)+\boldsymbol{\nabla}\lambda$$

Writing out the derivative and gradient of λ

$$\frac{\partial\lambda}{\partial t}=-\frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\boldsymbol{\nabla}\lambda=\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

gives the transformed potentials

$$V'(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$\mathbf{A}'(\mathbf{r},t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}} + \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2} = 0$$

$\mathbf{E}$ is now given by $-\boldsymbol{\nabla}V$ (instead of by $-\frac{\partial \mathbf{A}}{\partial t}$), while $\mathbf{B}$ is again 0, this time directly because $\boldsymbol{\nabla}\times\mathbf{A} = \boldsymbol{\nabla}\times 0 = 0$.

$$\mathbf{E} = -\boldsymbol{\nabla}V = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A} = 0$$

The result is exactly the same as in (a).

□

Problem 10.4

Suppose $V = 0$ and $\mathbf{A} = A_0\sin(kx-\omega t)\hat{\mathbf{y}}$, where $A_0, \omega,$ and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell's equations in vacuum. What conditions must you impose on ω and k?

Solution:

Maxwell's equations in vacuum are

$$\text{(i)}\boldsymbol{\nabla}\cdot\mathbf{E} = \frac{1}{\epsilon_0}\rho \qquad \text{(iii)}\boldsymbol{\nabla}\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$$

$$\text{(ii)}\boldsymbol{\nabla}\cdot\mathbf{B} = 0 \qquad \text{(iv)}\boldsymbol{\nabla}\times\mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}$$

where as before we set

$$\mathbf{E} = -\boldsymbol{\nabla}V - \frac{\partial\mathbf{A}}{\partial t}$$
$$\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A}$$

This gives

$$\mathbf{E} = -\frac{\partial\mathbf{A}}{\partial t} = -\omega A_0\cos(kx-\omega t)\hat{\mathbf{y}}$$

$$\mathbf{B} = \frac{\partial A_y}{\partial x}\hat{\mathbf{z}} = kA_0\cos(kx-\omega t)\hat{\mathbf{z}}$$

Since $\rho = 0$, this directly gives $\boldsymbol{\nabla}\cdot\mathbf{E} = 0$. (ii) gives $\boldsymbol{\nabla}\cdot\mathbf{B} = 0$.

From (iv) we have the relation, since the current density $\mathbf{J} = 0$,

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\downarrow$$

$$\boldsymbol{\nabla} \times \mathbf{B} = -\frac{\partial B_z}{\partial x}\hat{\mathbf{y}} = k^2 A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$

$$\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \epsilon_0 \omega^2 A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$

From this we see that $k^2 = \mu_0 \epsilon_0 \omega^2 \rightarrow k = \sqrt{\mu_0 \epsilon_0}\omega \rightarrow k = \frac{\omega}{c}$

From (iii) we have the relation

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\downarrow$$

$$\boldsymbol{\nabla} \times \mathbf{E} = \frac{\partial E_y}{\partial x}\hat{\mathbf{z}} = \omega k A_0 \sin(kx - \omega t)\hat{\mathbf{z}}$$

$$-\frac{\partial \mathbf{B}}{\partial t} = \omega k A_0 \sin(kx - \omega t)\hat{\mathbf{z}}$$

Equations (i)-(iv) are all verified with the condition $k = \frac{\omega}{c}$.

□

Problem 10.5

Which of the potentials

$$\text{(i) } V = 0, \quad \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\mathbf{z}}, & \text{for } |x| < ct \\ 0 & \text{for } |x| > ct \end{cases}$$

$$\text{(ii) } V = 0, \quad \mathbf{A} = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

$$\text{(iii) } V = 0, \quad \mathbf{A} = A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$

are in the Coulomb gauge? Which are in the Lorenz gauge?

Solution:

The Coulomb gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0$$

and the Lorenz gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Starting with checking the Coulomb gauge, we get

$$\text{(i) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(\frac{\mu_0 k}{4c} (ct - |x|)^2 \hat{\mathbf{z}} \right) = 0$$

$$\text{(ii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(-\frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{\mathbf{r}} \right) = -\frac{qt}{4\pi\epsilon_0} \boldsymbol{\nabla} \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = -\frac{qt}{4\pi\epsilon_0} \delta^3(\mathbf{r})$$

$$\text{(iii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot (A_0 \sin(kx - \omega t) \hat{\mathbf{y}}) = 0$$

(i) and (iii) are in the Coulomb gauge, but not (ii). Next, checking Lorenz gauge we see that the divergence of $\mathbf{A}$ should be proportional to the time derivative of V, but since $V = 0$ for (i) and (iii), they are also in the Lorenz gauge. This does not apply to (ii) since the divergence of $\mathbf{A} \neq 0$ while $\frac{\partial V}{\partial t} = 0$.

(i) Both Coulomb and Lorenz.

(ii) Neither Coulomb or Lorenz.

(iii) Both Coulomb and Lorenz. □

Problem 10.6

Problem 10.7

Problem 10.8

Problem 10.9

Problem 10.10

Confirm that the retarded potentials satisfy the Lorenz gauge condition. [Hint: First show that

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\mathscr{r}} \right) = \frac{1}{\mathscr{r}} (\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\mathscr{r}} (\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\mathscr{r}} \right)$$

where $\boldsymbol{\nabla}$ denotes derivatives with respect to $\mathbf{r}$, and $\boldsymbol{\nabla}'$ denotes derivatives with respect to $\mathbf{r}'$. Next, noting that $\mathbf{J}(\mathbf{r}', t - \mathscr{r}/c)$ depends on $\mathbf{r}'$ both explicitly and through $\mathscr{r}$, whereas it depends only on $\mathbf{r}$ through $\mathscr{r}$, confirm that

$$\boldsymbol{\nabla}\cdot\mathbf{J}=-\frac{1}{c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}\imath),\boldsymbol{\nabla}'\cdot\mathbf{J}=-\dot{\rho}-\frac{1}{c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}'\imath)$$

Use this to calculate the divergence of $\mathbf{A}$ (Eq. 10.26.)]

Solution:

We want to show

$$\boldsymbol{\nabla}\cdot\mathbf{A}=-\mu_0\epsilon_0\frac{\partial V}{\partial t}$$

where

$$\mathbf{A}=\frac{\mu_0}{4\pi}\int\frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath}\,\mathrm{d}\tau'$$

Now, taking the divergence of $\mathbf{A}$ gives

$$\boldsymbol{\nabla}\cdot\mathbf{A}=\frac{\mu_0}{4\pi}\int\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath}\right)\mathrm{d}\tau'$$

From now on, $\mathbf{J}(\mathbf{r}',t_r)=\mathbf{J}$. Using a divergence identity gives

$$\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}\cdot\mathbf{J})+\mathbf{J}\cdot\left(\boldsymbol{\nabla}\frac{1}{\imath}\right)$$

and

$$\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})+\mathbf{J}\cdot\left(\boldsymbol{\nabla}'\frac{1}{\imath}\right)$$

Since $\frac{1}{\imath}=\frac{1}{|\mathbf{r}-\mathbf{r}'|}$, we have

$$\boldsymbol{\nabla}'\frac{1}{\imath}=-\boldsymbol{\nabla}\frac{1}{\imath}$$

which means we can rewrite $\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)$ as

$$\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})-\mathbf{J}\cdot\left(\boldsymbol{\nabla}\frac{1}{\imath}\right)$$

$$\downarrow$$

$$\mathbf{J}\cdot\left(\boldsymbol{\nabla}\frac{1}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})-\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)$$

Putting this expression into $\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}}{\imath}\right)$ gives

$$\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}\cdot\mathbf{J})+\frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})-\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)$$

which is what the hint wanted us to show. Putting this into the expression for $\boldsymbol{\nabla}\cdot\mathbf{A}$ gives

$$\boldsymbol{\nabla}\cdot\mathbf{A}=\frac{\mu_0}{4\pi}\int\left[\frac{1}{\imath}(\boldsymbol{\nabla}\cdot\mathbf{J})+\frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})-\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)\right]\mathrm{d}\tau' \tag{1}$$

According to the Lorenz gauge condition,

$$\boldsymbol{\nabla}\cdot\mathbf{A}=-\mu_0\epsilon_0\frac{\partial V}{\partial t}=-\frac{\mu_0}{4\pi}\int\frac{\dot{\rho}}{\imath}\,\mathrm{d}\tau'$$

With this information, we now evaluate the three terms in Equation 1

Starting with the first term, and using the chain rule on $\mathbf{J}$. Since $\mathbf{J}$ depends on $\mathbf{r}$ only through t_r,

$$\begin{gathered}\boldsymbol{\nabla}\cdot\mathbf{J}=\dot{\mathbf{J}}\cdot\boldsymbol{\nabla}t_r=-\frac{1}{c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}\imath)\\ \downarrow\\ \frac{1}{\imath}(\boldsymbol{\nabla}\cdot\mathbf{J})=-\frac{1}{\imath c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}\imath)\end{gathered} \tag{2}$$

The next term is the divergence of $\mathbf{J}$ with respect to $\mathbf{r}'$. $\mathbf{J}$ depends on $\mathbf{r}'$ both explicitly and through t_r, so

$$\begin{gathered}\boldsymbol{\nabla}'\cdot\mathbf{J}=(\boldsymbol{\nabla}'\cdot\mathbf{J})_{t_r}+\dot{\mathbf{J}}\cdot\boldsymbol{\nabla}'t_r=-\dot{\rho}-\frac{1}{c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}'\imath)\\ \downarrow\\ \frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})=-\frac{\dot{\rho}}{\imath}-\frac{1}{\imath c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}'\imath)=-\frac{\dot{\rho}}{\imath}+\frac{1}{\imath c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}\imath)\end{gathered} \tag{3}$$

where $(\boldsymbol{\nabla}'\cdot\mathbf{J})_{t_r}$ is the divergence taken at fixed t_r, which equals $-\dot{\rho}$ by the continuity equation.

The last term integrates to 0 which I will explain later.

Adding Equation 2 and Equation 3, the only term left is

$$-\frac{\dot{\rho}}{\imath}$$

which gives

$$\begin{gathered}\boldsymbol{\nabla}\cdot\mathbf{A}=\frac{\mu_0}{4\pi}\int\left(-\frac{\dot{\rho}}{\imath}\right)\mathrm{d}\tau'=-\frac{\mu_0\epsilon_0}{4\pi\epsilon_0}\int\frac{\dot{\rho}}{\imath}\,\mathrm{d}\tau'=\\ =-\mu_0\epsilon_0\frac{\partial}{\partial t}\left[\frac{1}{4\pi\epsilon_0}\int\frac{\rho}{\imath}\,\mathrm{d}\tau'\right]=-\mu_0\epsilon_0\frac{\partial V}{\partial t}\\ \downarrow\\ \boxed{\boldsymbol{\nabla}\cdot\mathbf{A}=-\mu_0\epsilon_0\frac{\partial V}{\partial t}}\end{gathered}$$

□

Problem 10.11

Problem 10.12

Problem 10.13

Solution:

We have

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t_r)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} + \frac{\dot{\rho}(\mathbf{r}',t_r)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} - \frac{\dot{\mathbf{J}}(\mathbf{r}',t_r)}{c^2\mathscr{r}}\right]\mathrm{d}\tau' \tag{4}$$

and

$$\rho(\mathbf{r},t) = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)t$$

$$\rho(\mathbf{r},t_r) = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)t_r = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)\left(t - \frac{\mathscr{r}}{c}\right)$$

$$\downarrow$$

$$\rho(\mathbf{r},t_r) = \rho(\mathbf{r},t) - \dot{\rho}(\mathbf{r},0)\frac{\mathscr{r}}{c}$$

$$\downarrow$$

$$\frac{\rho(\mathbf{r},t_r)}{\mathscr{r}^2} = \frac{\rho(\mathbf{r},t)}{\mathscr{r}^2} - \frac{\dot{\rho}(\mathbf{r},0)}{c\mathscr{r}}$$

Putting this into Equation 4, and noting that $\dot{\mathbf{J}} = 0$ since $\mathbf{J}$ is constant in time, we get

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} - \frac{\dot{\rho}(\mathbf{r}',0)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} + \frac{\dot{\rho}(\mathbf{r}',t_r)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}}\right]\mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{J} \text{ constant in time} \leftrightarrow \dot{\rho} \text{ constant in time } [\boldsymbol{\nabla}\cdot\mathbf{J} = -\dot{\rho}] \rightarrow \dot{\rho}(\mathbf{r},t) = \dot{\rho}(\mathbf{r},0)$$

$$\downarrow$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} - \frac{\dot{\rho}(\mathbf{r}',0)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} + \frac{\dot{\rho}(\mathbf{r}',0)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}}\right]\mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\frac{\rho(\mathbf{r}',t)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}\,\mathrm{d}\tau'$$

□

Problem 10.14

Show that the Biot-Savart law holds, with $\mathbf{J}$ evaluated at the nonretarded time, considering only the first two terms in the Taylor expansion for $\mathbf{J}$ is

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t) + ...$$

where the $\mathbf{r}$ dependence in $\mathbf{J}(\mathbf{r}, t_r)$ has been omitted for compactness.

Solution:

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{\mathscr{r}^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c\mathscr{r}} \right] \times \hat{\boldsymbol{\mathscr{r}}} \, \mathrm{d}\tau' \tag{5}$$

We use the given Taylor expansion on the first term in Equation 5:

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t)$$

$$t_r = t - \frac{\mathscr{r}}{c} \rightarrow t_r - t = -\frac{\mathscr{r}}{c}$$

$$\downarrow$$

$$\mathbf{J}(t_r) = \mathbf{J}(t) - \frac{\mathscr{r}}{c}\dot{\mathbf{J}}(t)$$

$$\downarrow$$

$$\frac{\mathbf{J}(t_r)}{\mathscr{r}^2} = \frac{\mathbf{J}(t)}{\mathscr{r}^2} - \frac{\dot{\mathbf{J}}(t)}{c\mathscr{r}} \tag{6}$$

and on the second term of Equation 5:

$$\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t) + (t_r - t)\ddot{\mathbf{J}}(t)$$

But since all higher order terms are neglected, $\ddot{\mathbf{J}}(t) = 0$ so

$$\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t) \tag{7}$$

Putting Equation 6 and Equation 7 into Equation 5 gives

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t)}{\mathscr{r}^2} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t)}{c\mathscr{r}} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t)}{c\mathscr{r}} \right] \times \hat{\boldsymbol{\mathscr{r}}} \, \mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t)}{\mathscr{r}^2} \times \hat{\boldsymbol{\mathscr{r}}} \, \mathrm{d}\tau'$$

□

Problem 10.15

Problem 10.16

Problem 10.17

Problem 10.18

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